



UNPOISON
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PUBLIC COMMENT

Comments on Draft National Biodiversity Strategy and Action Plan 2026-2035

Reversing pesticide and nutrient pollution driven biodiversity loss, and restoring South Africa's ecosystems, wildlife, soils, and waters.

*The Strategy requires **measurable reductions** in pesticide risk, nutrient loss, wastewater pollution and modern chemical contamination across **agriculture, cities, transport corridors, conservation, invasive alien species management, forestry and water.***

Date: 9 September 2026

Notice: Government Gazette No. 55141, Government Notice 7773 of 2026

Submitted to: Department of Forestry, Fisheries and the Environment

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Central request. Revise the NBSAP so that it contains explicit, time-bound and measurable commitments for national pathways to halve pesticide and their metabolites risk and excess nutrient losses by 2030; progressively phase out highly hazardous pesticides; require non-chemical-first, transparently monitored pest and alien invasive species management across all public and private sectors; and modernise source-to-tap water-quality, source-to-sea wastewater and coastal-discharge governance through explicit DWS - DFFE accountability, stronger Green Drop and enforceable effluent and ecological requirements, and risk-based source control, effects-based monitoring of pharmaceuticals and their transformation products, PFAS, antibiotics, antimicrobial-resistance indicators and all contaminants of emerging concern and other priority toxic pollutants.

1. Executive summary

UnPoison welcomes the draft NBSAP's dedicated thematic area on pollution and waste, its express recognition of pesticide impacts, its One Health action, and its proposed support for an HHP registry and substitution with safer alternatives. These are important advances. In its present form, however, the NBSAP does not yet provide a credible national implementation pathway for the pesticide and nutrient elements of the Kunming-Montreal Global Biodiversity Framework (KMGBF).

Most critically, KMGBF Target 7 requires action by 2030 and contains two quantitative commitments: at least a 50% reduction in excess nutrients lost to the environment, and at least a 50% reduction in the overall risk from pesticides and highly hazardous chemicals. The draft shifts the deadline to 2035 and replaces both requirements with an undefined 'decrease'. It also confines the only pesticide-specific action to research on agricultural systems, leaving pesticide use in cities, forestry, conservation estates, protected areas, invasive alien species clearing and other public programmes outside the implementation architecture.

In addition, its treatment of wastewater and coastal pollution does not expressly assign DWS responsibility alongside DFFE, strengthen Green Drop and associated discharge requirements, or establish a coordinated framework for contaminants of emerging concern and other priority toxic pollutants. These include pharmaceuticals, personal-care products, their metabolites and transformation products; endocrine-disrupting chemicals; PFAS; antibiotics, resistant organisms and antimicrobial-resistance genes; micro- and nanoplastics; pesticides; industrial chemicals; and other trace organic and inorganic pollutants. Within the connected water cycle, these contaminants may enter wastewater systems and receiving waters, persist or transform during treatment, accumulate in sludge, biosolids and sediments, and reach aquatic ecosystems, groundwater and drinking-water sources, creating material biodiversity, water-security and One Health risks.

UnPoison requests eight essential amendments:

- **Restore the 2030 deadline and the two 'at least half' commitments** in the national target and its indicators.
- **Create a transparent HHP phase-out pathway** with annual withdrawal milestones, no indefinite derogations, and an ultimate objective of complete phase-out.
- **Add a national nutrient-loss programme** addressing synthetic nitrogen and phosphorus fertilisers, nutrient-use efficiency, nitrate and phosphate pollution, and rapid adoption of lower-impact soil-fertility systems.
- **Extend pesticide governance beyond agriculture** to urban public space, households and retail, forestry, conservation, protected areas, transport corridors, water infrastructure and invasive alien species management.
- **Require a non-chemical-pesticide-first control hierarchy** for all state-funded biodiversity, restoration and alien and invasive species clearing programmes, with HHPs prohibited in these programmes on an accelerated timetable.
- **Establish public use and environmental monitoring** covering active ingredients, relevant co-formulants, metabolites, quantities, transformation products, location, purpose, toxicity-weighted risk, environmental concentrations and non-target effects.
- **Assign joint responsibility and accountability** to DFFE, the Department of Agriculture, DWS, SANBI, SANParks, the Department of Transport and relevant road- and rail-infrastructure authorities, provincial conservation authorities, COGTA/SALGA, and municipalities, with civil-society participation.
- **Modernise wastewater and coastal-discharge governance** by assigning explicit responsibilities to DWS alongside DFFE and local government; strengthening Green Drop criteria and discharge authorisations; embedding a coordinated national framework for

contaminants of emerging concern and other priority toxic pollutants that introduces risk-based monitoring and source control for pharmaceuticals, their transformation products, antibiotics and antimicrobial resistance indicators, PFAS and all the contaminants of emerging concern, most notably EDCs, as well as permit review and public reporting.

Decision requested: *Do not defer these core targets to later annual work plans, the National Biodiversity Framework or sector strategies. The national target, timeframes, headline indicators and assigned mandates must be explicit in the final NBSAP itself.*

2. About UnPoison and our interest in the NBSAP

UnPoison is a multi-sector strategic expert network and registered South African advocacy NPO working to reform the regulatory framework governing agrochemicals, contaminants and pollution governance, to phase out highly hazardous pesticides (HHPs), improve public access to hazardous substances information and to develop solutions to accelerate a just transition to safer alternatives. Our work sits at the intersection of biodiversity, water, food systems, public health, labour rights, community participation and One Health.

We submit from both extensive regulatory and on-the-ground experience.

UnPoison is available to support the defining and implementation of a gold standard biodiversity strategy for South Africa with research, convenings, tools, systems architecture and access to its high-level expert network to help South Africa reach its goals on time and mitigate compounding ecosystem degradation and biodiversity loss.

3. What the draft gets right

- A dedicated pollution and waste management thematic area that corrects a major omission in the previous NBSAP.
- The rationale recognises agricultural nutrient runoff, hazardous substances and the direct and indirect effects of pesticides on pollinators and other biodiversity.
- The draft proposes an HHP registry, an HHP policy and substitution with safer alternatives.
- The mainstreaming section calls for a One Health approach across environment, agriculture and health.
- The draft proposes a pesticide environmental concentration and/or aggregated total applied toxicity indicator, which is more meaningful than measuring kilograms alone.
- The draft recognises the need to connect pollution control, ecosystem restoration, water quality, transparent information and community participation.

These foundations should be retained and strengthened. Our submission is directed at turning them into an implementable national pathway rather than removing or diluting them.

4. Material defects in the present draft

4.1 The national pollution target is weaker than KMGBF Target 7

The NBSAP states that section 5.7 is aligned to KMGBF Target 7. Yet the global target is expressly a 2030 target and requires reducing excess nutrients lost to the environment by at least half and reducing overall risk from pesticides and highly hazardous chemicals by at least half. The draft national target instead seeks a general reduction to non-harmful levels by 2035. Its pesticide indicator requires only a 'decrease', and it contains no quantitative nutrient-loss indicator at all.

Extending the deadline and replacing the KMGBF's quantified reduction commitments with an undefined requirement to "decrease" pesticide risk, and no quantitative nutrient-loss indicator, cannot be regarded as alignment with Target 7.

The absence of a current baseline is not a justification for omitting the target: the NBSAP can require a baseline to be established urgently while retaining the internationally agreed outcome and deadline.

Required correction: *Retain 2035 as the outer horizon of the NBSAP, but insert explicit and measurable commitments as 2027 baseline milestones and the full 2030 Target 7 outcomes, followed by post-2030 consolidation and complete HHP phase-out milestones.*

4.2 Research is not a phase-out or implementation strategy

The draft's only pesticide-specific action is to conduct assessments and research on pesticide use in agricultural systems to support the HHP policy, HHP registry and substitution. Research is necessary, but South Africa already has sufficient knowledge to begin preventive action. The NBSAP's own precautionary and One Health orientation means uncertainty must shape monitoring and safeguards; it must not postpone withdrawal of the most hazardous products, risk reduction or deployment of established alternatives.

The phrase 'safer alternatives' also needs a clear hierarchy. Substitution must not default to another synthetic pesticide with similar chronic, ecological or cumulative risks.

Prevention, manual and mechanical controls, ecological restoration, biological control, agroecology, cultural practices, and properly regulated biological products **must** be considered before lower-hazard chemical options.

4.3 Fertiliser pollution is acknowledged but not governed

Section 5.7 refers to agricultural nutrient runoff, nutrient loading, coastal eutrophication and implementation of the National Eutrophication Management Strategy. It nevertheless **contains no quantified reduction target for agricultural nitrogen or phosphorus losses, no national nutrient balance, no fertiliser-use indicator, no action directed at fertiliser practices or products, and no assigned Department of Agriculture role in the relevant pollution actions.**

The correct objective is not an indiscriminate reduction in nutrients needed for food production. It is to halve avoidable nutrient losses, progressively phase down high-loss and demonstrably harmful fertiliser practices and products, improve nutrient cycling and support farmers to adopt precision application, soil testing, legume rotations, composts and organic amendments, biofertilisers and other regenerative soil-health approaches.

4.4 The draft excludes major non-agricultural pesticide pathways

Pesticides are not only used in agriculture. They are used in forestry, across national and urban roads, verges, parks, sports grounds, schools, stormwater systems, industrial sites, rail and utility corridors, plantations, protected areas, conservation estates and invasive-alien vegetation programmes. These applications can occur in or directly beside critical biodiversity areas, rivers, wetlands, estuaries, strategic water source areas, pollinator habitat and residential communities. They are not captured by the agriculture-only research action.

This glaring gap, including the urban context, is a dangerous threat to South Africa's megabiodiversity. Section 5.1 measures the share of green and blue public space but does not require that these spaces be managed in a biodiversity-safe manner. A chemically maintained verge or park should not qualify as a positive biodiversity outcome because it is mapped as green space. **Chemically maintained areas should be removed from the green space as they do not qualify as green space as most chemicals have hazard risks and contribute towards greenhouse gas emissions.**

An associated gap is pesticide-related wildlife poisoning. Pesticides are deliberately used in baits to kill predators in livestock and wildlife areas, while vultures and other scavengers may be killed when they feed on the bait or poisoned carcasses. Rodenticides result in widespread secondary poisoning of owls, raptors and mammalian carnivores and predators in rural conservation and urban environments.

Cape Town illustrates the scale and transparency gap: the City reportedly serviced 63,997 and 44,121 rat bait-stations in consecutive comparable nine-month periods, while a 2026 study detected anticoagulant rodenticide residues in 78.4% of 102 caracals, (although it did explicitly attribute those exposures to municipal operations).

No standardised national dataset of municipal rodenticide use and non-target effects is publicly available. These pathways require product controls, poisoning-incident surveillance, response and enforcement, non-lethal predator-conflict mitigation, waste- and non-chemical-first rodent control, and transparent municipal pesticide-use reporting.

The institutional omission is equally important. **National, provincial and municipal road authorities and rail-infrastructure managers** procure and oversee vegetation control across extensive linear corridors, but the draft assigns them no responsibility for pesticide-risk reduction, HHP exclusion, public use reporting or biodiversity-safe vegetation management. **The relevant entities include the Department of Transport, SANRAL, provincial road authorities, municipalities, Transnet Rail Infrastructure Manager and PRASA**, together with concessionaires and contractors acting on their behalf.

4.5 Alien-species control can create a second biodiversity pressure

Section 5.2 sets substantial hectare targets for invasive-alien clearing and restoration and promotes classical biological control, but no safeguard governing herbicide selection or use. Conservation and alien-clearing programmes themselves use hazardous herbicides, including paraquat in some conservation contexts. Without a control hierarchy, transparent chemical-use data and non-target monitoring, a programme may report hectares cleared while transferring harm to soil organisms, insects, indigenous plants, aquatic life, workers or downstream water users.

Restoration success must therefore be measured by ecological recovery and reduced chemical burden, not only by hectares treated. Chemical control, where demonstrably necessary, must be targeted, time-limited, legally registered for the use, independently monitored and excluded from sensitive buffers and HHP categories.

4.6 Monitoring is too narrow to support accountability

Environmental concentration is essential but cannot stand alone. It may miss episodic exposures, spray drift, short-lived but acutely toxic compounds, metabolites, formulation effects and biodiversity harm caused below routine detection thresholds. Nor can kilograms applied alone reflect hazard. The NBSAP needs a linked set of pressure, risk, concentration and outcome indicators, disaggregated by active ingredient, relevant co-formulants, sector, place and receiving ecosystem.

Monitoring must cover the relevant environmental media, including surface water, groundwater, soil, sediment, air or spray deposition where relevant, and biota, and combine routine monitoring with event-based sampling following spraying, rainfall, spills, bypasses and other pollution incidents.

Contaminants of emerging concern comprise evolving groups and complex mixtures that cannot be managed through a fixed analyte list alone, so monitoring should also combine priority-target analysis with non-target screening, effects-based bioassays and whole-effluent toxicity testing.

4.7 The institutional table understates the Department of Agriculture's mandate

The draft's legislation table correctly lists the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act 36 of 1947. However, section 2.2.4's table of departmental roles omits Act 36 from the Department of Agriculture's relevant legislation, even though pesticide and fertiliser regulation are central to the proposed pollution target. This should be corrected, and the Department of Agriculture should be expressly assigned pesticide-risk, HHP phase-out, biological-alternative and nutrient-loss responsibilities.

4.8 Wastewater, contaminants of emerging concern and other priority toxic pollutants are inadequately governed

The draft NBSAP's page 79 action on marine pollution assigns responsibility to DFFE Oceans and Coasts but does not expressly include DWS, COGTA/SALGA or municipal wastewater authorities. The final NBSAP should distinguish dumping at sea and ship-based discharges from land-based wastewater and industrial effluent entering coastal waters. DFFE should retain responsibility for coastal waters discharge and dumping permits, while DWS should be expressly responsible for wastewater-system regulation, Green Drop oversight, National Water Act controls and upstream catchment and estuarine water quality.

The pollutant coverage is also incomplete. Contaminants of emerging concern and other priority toxic pollutants enter the connected water cycle from municipal, healthcare, household, agricultural, mining and industrial sources. Relevant groups include pharmaceuticals, their biologically active metabolites and transformation products; personal-care products; endocrine-disrupting chemicals; PFAS; antibiotics, resistant organisms and antimicrobial-resistance genes; micro- and nanoplastics; pesticides; industrial chemicals; and other trace organic and inorganic pollutants. Established hazards such as HHPs and toxic metals, including mercury, lead and cadmium, must be controlled alongside CECs without being misclassified as novel pollutants.

Depending on their properties and the treatment processes applied, these contaminants may persist or transform during wastewater treatment, accumulate in sludge, biosolids or sediments, or reach receiving waters, groundwater, aquatic ecosystems and drinking-water sources. The WRC, in collaboration with DWS, has identified fragmented monitoring, inadequate priority lists, limited harmonisation of analytical methods, uneven laboratory capacity, insufficient risk guidance and weak integration of evidence into policy and regulation, and is piloting a coordinated national water programme on CECs. The NBSAP should embed this programme within Target 7 through prevention and source control; dynamic priority and watch lists; coordinated monitoring of wastewater, sludge and biosolids, water columns, sediments, groundwater and drinking-water sources; targeted chemical and effects-based methods; treatment and permit review; and transparent public reporting.

SANS 241 is a drinking-water standard and cannot substitute for enforceable effluent, receiving-water and ecological criteria. Green Drop criteria and associated discharge authorisations should therefore be strengthened to address priority modern synthetic pollutants, cumulative and mixture effects, receiving-ecosystem impacts and, at appropriate high-risk facilities, whole-effluent toxicity.

5. Draft-ready amendments

The following wording is offered so that the requested changes can be incorporated directly without requiring a redesign of the NBSAP's architecture.

5.1 Section 2.2.4 - Institutional context

Replace or expand the Department of Agriculture mandate: Sustainable agriculture and agro-biodiversity conservation; regulation of pesticides, fertilisers, biological inputs and other agricultural remedies across their life cycles; reduction of pesticide risk and excess nutrient losses; implementation of the national HHP phase-out pathway and transparent pesticide registry; support for integrated pest management, agroecology and safer biological and non-chemical alternatives; plant and animal health; and food-system resilience. Relevant instruments include the Fertilizers, Farm Feeds, Agricultural Remedies and Stock Remedies Act 36 of 1947, the Conservation of Agricultural Resources Act 43 of 1983, the Preservation and Development of Agricultural Land Act 39 of 2024, and the Pesticide Policy Framework and associated implementation instruments.

Expand DFFE's mandate: Biodiversity impact assessment and monitoring of pesticides, fertilisers and other harmful contaminants across terrestrial, freshwater, estuarine and marine realms; safeguards for pesticide use in protected areas, restoration, invasive-species management and environmental programmes; cross-departmental coordination under NEMA and the Biodiversity Act; and public reporting toward KMGBF Target 7.

Expand DWS's mandate: Regulation and enforcement of municipal and industrial wastewater discharges under the National Water Act; oversight and strengthening of the Green Drop programme; establishment of catchment, estuarine and receiving-water quality objectives; alignment with the national WRC-DWS programme on contaminants of emerging concern; and modernisation of discharge conditions and monitoring to address pesticides, nutrients, CECs and other priority toxic pollutants. DWS must act jointly with DFFE Oceans and Coasts where land-based effluent enters coastal waters and must publish facility-level compliance, corrective-action and enforcement information.

Add Department of Transport and transport-infrastructure responsibilities: Integrate pesticide-risk reduction, invasive-species obligations and biodiversity-safe vegetation management into national road and rail policy, engineering standards, public-entity oversight, procurement, concession and maintenance contracts. SANRAL, provincial road authorities, municipalities, Transnet Rail Infrastructure Manager, PRASA and other relevant infrastructure managers must implement these requirements on land under their control, retain accountability for concessionaire and contractor performance, and publish pesticide-use, incident and ecological-outcome information.

Expand municipal and COGTA/SALGA responsibilities: Integration of pesticide and fertiliser reduction into municipal biodiversity by-laws, Integrated Development Plans, green and blue space management, parks and road-verge operations, stormwater management, procurement, public notification, annual reporting, operation and maintenance of wastewater systems; industrial and trade-effluent pre-treatment and by-laws; sewage spill and bypass reporting; pharmaceutical take-back arrangements; and public disclosure of effluent and receiving-water monitoring.

5.2 Section 5.1 - Mainstreaming biodiversity

Add the following indicators and actions:

- Define 'productive and sustainable agriculture' so that qualifying areas must demonstrate measurable pesticide-risk reduction, HHP withdrawal, improved nutrient-use efficiency, protection of soil and aquatic biodiversity, and adoption of agroecological, IPM and biological alternatives. Certification alone must not be treated as conclusive evidence.
- By 2028, 100% of municipalities have adopted and begun implementing a pesticide-reduction and biodiversity-safe green/blue-space management plan, including an HHP prohibition for public spaces, public notification and annual use reporting.
- Incorporate pesticide and nutrient criteria into the model municipal biodiversity by-law, municipal biodiversity indicators, IDP reporting and the District Development Model.

- Require large companies in agriculture, forestry, landscaping, food retail and pesticide manufacture to disclose pesticide risk, HHP use, nutrient losses and time-bound transition plans as part of biodiversity dependencies and impacts.
- Reform incentives and public procurement so that state funding supports manual, mechanical, ecological and biological control, agroecology and nutrient efficiency, and does not subsidise or procure HHPs except under narrowly defined, transparent and time-limited essential-use exemptions.

5.3 Section 5.2 - Restoration and invasive alien species

Insert in the rationale: *Control methods can themselves create biodiversity, water-quality and occupational-health risks. Restoration and invasive-species management must therefore apply an intervention hierarchy that prioritises prevention, ecological, manual, mechanical and appropriately regulated biological control; uses chemical control only where demonstrably necessary; excludes HHPs on an accelerated timetable; and monitors non-target effects, chemical residues, worker and community exposure, water quality and indigenous ecological recovery.*

Add target indicator: *By 2028, 100% of state-funded, state-managed and protected-area invasive-species and restoration programmes implement a published non-chemical-first pesticide management standard and report pesticide use annually by active ingredient, formulation, quantity, site, date, target species and method; by 2030, no HHP is used in these programmes except under a published, independently reviewed, time-limited essential-use exemption.*

Add action: *Develop and implement binding national minimum standards for chemical use in restoration and invasive-species management under NISSAP, Working for Water and protected-area management. The standards must require an alternatives assessment, prohibit broadcast application in sensitive areas, establish no-spray buffers for water bodies and priority biodiversity areas, regulate contractors, prevent drift and runoff, require public records and incident reporting, and measure restoration success through indigenous recovery and ecosystem function rather than hectares treated alone. Lead: DFFE. Support: SANBI, SANParks, DWS, DoA, provincial conservation agencies, SALGA, municipalities, research institutions, workers and civil society.*

5.4 Section 5.7 - Pollution and waste management

Replace the national target: *By 2030, reduce pollution risks and negative impacts from all sources to levels not harmful to people, biodiversity, ecosystem functions and services, considering cumulative and mixture effects, including by reducing excess nitrogen and phosphorus lost to the environment by at least half through more efficient nutrient cycling and use; reducing the overall risk from pesticides and highly hazardous chemicals by at least half through prevention, agroecology, integrated pest management and safer biological and non-chemical alternatives; preventing, reducing and working towards eliminating plastic pollution, including micro- and nanoplastics; and implementing a time-bound pathway to phase out highly hazardous pesticides. By 2035, consolidate these reductions, complete the phase-out of HHPs except for narrowly defined and time-limited essential uses, and maintain pollution below ecological harm thresholds.*

Proposed target indicators

- **Baseline:** By the end of 2027, establish and publish national and provincial baselines for pesticide sales and use, HHP use, aggregated total applied toxicity, pesticide environmental concentrations, excess nitrogen and phosphorus losses, nutrient balances and fertiliser use, disaggregated by sector and receiving ecosystem.
- **Pesticide risk:** By 2030, reduce the national aggregated risk from pesticides and highly hazardous chemicals by at least 50% from the published baseline, with results reported by active ingredient, hazard class, use sector and province.
- **HHP phase-out:** Publish an annual HHP withdrawal schedule and measure the number, volume and uses of HHPs removed; prohibit new registrations and new uses of HHPs; end

open-ended derogations; and by 2035 phase out remaining HHPs except narrowly defined, published, independently reviewed and time-limited essential uses with substitution plans.

- **Nutrient loss:** By 2030, reduce excess nitrogen and phosphorus lost to the environment by at least 50%, measured through national and catchment nutrient balances, reactive nitrogen loss, nitrogen deposition, nitrate and phosphate in surface and groundwater, agricultural return flows and coastal eutrophication.
- **Contaminants of emerging concern and other priority toxic pollutants:** By the end of 2028, align NBSAP implementation with the national WRC-DWS CEC programme by publishing dynamic, risk-based national priority and watch lists and a baseline covering pharmaceuticals and transformation products, endocrine-disrupting chemicals, PFAS, antibiotics and antimicrobial-resistance determinants, micro- and nanoplastics, pesticides and priority toxic metals in wastewater effluent, sludge and biosolids, sediments and priority receiving waters, groundwater and drinking-water sources. By 2030, review all coastal and other high-risk wastewater discharge authorisations against updated requirements and publish trends in concentrations, loads, effects-based toxicity, treatment performance and ecological effects.
- **Public-sector transition:** By 2028, all organs of state report pesticide use and implement non-chemical-first IPM plans; by 2030, no HHP is procured or used in public urban, conservation, restoration, roadside or water-management programmes except under the essential-use process.
- **Biodiversity outcomes:** Track trends in pollinators, soil biodiversity, aquatic macroinvertebrates, birds and other appropriate sentinel taxa in pesticide- and nutrient-pressure hotspots, linked to use and concentration data.
- **Alternative uptake:** Increase annually the proportion of agricultural, forestry, municipal and conservation land under verified agroecological, IPM, biological, manual and mechanical pest or vegetation management, while maintaining food security, livelihoods and ecological recovery.

Required amendments and additions to existing section 5.7 actions on draft page 79

Add a transport-corridor vegetation-management action: Require SANRAL, provincial road authorities, Transnet Rail Infrastructure Manager, PRASA and other relevant road and rail infrastructure managers to develop and implement binding minimum standards for vegetation and pesticide management across national and provincial road reserves and rail corridors, including all work undertaken by concessionaires and contractors.

The standards must distinguish safety-critical clearance zones from ecological verge and buffer zones; require site-specific planning and documented consideration of manual, mechanical, ecological and biological alternatives before chemical control; permit targeted chemical use only where demonstrably necessary for operational safety or invasive-species control; progressively exclude HHPs in accordance with the national phase-out timetable; require legally registered products and appropriately qualified operators; protect watercourses, wetlands, strategic water source areas, critical biodiversity areas, protected species, pollinator habitat and neighbouring communities; and require georeferenced pesticide-use records, incident reporting, advance public notification where exposure may occur, annual public disclosure and monitoring of ecological outcomes.

Lead: Department of Transport, with DFFE and the Department of Agriculture acting within their respective statutory mandates.

Support: DWS, SANBI, COGTA/SALGA, municipalities, the Railway Safety Regulator, research institutions and civil society.

Replace or expand the page 79 marine-pollution action: Strengthen regulation and permitting for all sources of marine pollution, distinguishing dumping at sea and ship-based discharges from land-based municipal and industrial effluent entering coastal waters. Review and enforce discharge conditions, cumulative-load limits, receiving-water objectives and ecological thresholds; include nutrients, pesticides, contaminants of emerging concern and other priority toxic pollutants, including pharmaceuticals and their metabolites and transformation products, endocrine-disrupting chemicals, PFAS, antibiotics and antimicrobial-resistance determinants, micro- and nanoplastics and priority toxic metals; and require source control, industrial pre-treatment, risk-based treatment upgrades, whole-effluent toxicity testing at appropriate high-risk facilities and public reporting.

Lead: DFFE Oceans and Coasts and DWS, within their respective statutory mandates.

Support: COGTA/SALGA, municipalities, DoH/SAHPRA, Department of Agriculture, SANBI, WRC, research institutions and civil society.

Strengthen Green Drop and linked discharge controls: By 2028, DWS must review and strengthen Green Drop criteria and the discharge authorisations and effluent requirements against which wastewater systems are assessed. The revised framework must address sewage spills and bypasses, facility-level pollutant loads, receiving-water and ecological outcomes, sludge and biosolids, cumulative and mixture effects, and risk-based monitoring of contaminants of emerging concern and other priority toxic pollutants, using targeted chemical analysis, effects-based methods and whole-effluent toxicity testing where appropriate. Annual plant-level results, corrective-action plans, deadlines and enforcement status must be publicly accessible. **Lead:** DWS. **Support:** DFFE, COGTA/SALGA, municipalities, DoH/SAHPRA, WRC and SANBI.

Proposed actions

1. **Create a national pesticide-risk and HHP phase-out implementation plan** under joint DFFE and Department of Agriculture leadership, with annual milestones, public reporting, an independent scientific and civil-society advisory mechanism, and no indefinite derogations.
2. **Establish mandatory pesticide-use reporting** for registrants, distributors, commercial users and organs of state, covering active ingredients, relevant co-formulants, formulation, quantity, crop or target, location, date and method. Publish aggregated data in open formats while protecting legitimate personal information, not regulatory or environmental facts.

3. **Integrate biodiversity into Act 36 decision-making** through full-formulation and co-formulant assessment, chronic and endocrine endpoints, pollinator and soil-organism effects, environmental fate, mixtures, cumulative exposure and South African ecosystem vulnerability; link DFFE and DWS evidence directly to registration, re-evaluation and withdrawal decisions.
4. **Adopt a substitution hierarchy** that prioritises prevention, cultural and ecological controls, agroecology, manual and mechanical methods, safe biological control and properly regulated biological products across all pesticide uses, before lower-hazard chemical options, and requires assessment to prevent regrettable substitution.
5. **Develop a national sustainable nutrient management programme** led jointly by DWS and the Department of Agriculture, with catchment-specific actions, farmer support, nutrient budgeting, soil testing, precision application, improved nutrient cycling, biofertilisers, organic amendments and progressive phase-down of high-loss fertiliser practices and products.
6. **Protect water and ecological infrastructure** through risk based source control and coordinated, routine and event-based monitoring of pesticides and metabolites, nutrients, pharmaceuticals and their transformation products, PFAS, antibiotics and antimicrobial-resistance indicators in strategic water source areas, groundwater vulnerability zones, rivers, wetlands, estuaries, agricultural return flows and urban stormwater, using chemical analysis, effects-based methods and event-based sampling where appropriate e.g. after rainfall and herbicide spraying.
7. **Adopt national minimum municipal standards** for parks, road verges, stormwater, schools, sports grounds and other public spaces, including advance public notification, sensitive-site buffers, an HHP prohibition, contractor controls, manual-weeding and ecological-management budgets, rodent control led by waste prevention and non-chemical methods with safeguards against secondary poisoning, incident reporting and standardised annual public disclosure of municipal pesticide use, including rodenticides.
8. **Adopt national minimum conservation and forestry standards** for protected areas, biodiversity stewardship, plantation forestry, restoration and invasive-species control, including the safeguards proposed under section 5.2 and separate reporting of pesticide performance rather than reliance on certification status alone.
9. **Fund transition and implementation** by redirecting harmful incentives and procurement, supporting farmer and municipal transition costs, strengthening public extension and registration capacity for biological alternatives, and creating jobs in manual, mechanical, ecological and biological control.
10. **Ensure meaningful participation and accountability** by including farm workers, affected communities, municipalities and municipal workers, conservation and wildlife organisations, independent scientists and civil society in indicator design, hotspot monitoring, annual review and phase-out decisions, and by providing accessible incident and environmental data.

5.5 Amend the section 5.7 theory of change

Replace the IF statement: *IF environmental, agricultural, water, health, labour, forestry and local-government mandates are harmonised; pesticide risk and excess nutrient losses are baselined and reduced against time-bound targets; HHPs are progressively phased out; and affected people have transparent information and meaningful participation,*

Replace the THEN statement: *THEN pollution from pesticides, fertilisers and other major sources will decline measurably across terrestrial, freshwater, estuarine and marine ecosystems, while safer pest management, resilient production and ecological restoration expand,*

Replace the BECAUSE statement: *BECAUSE coordinated regulation, public data, precautionary withdrawal of the most hazardous substances, support for alternatives, independent monitoring and accountability convert broad policy intent into prevention at source and measurable biodiversity recovery.*

5.6 Sections 5.3 and 5.8 - Ecosystem conservation and ecological infrastructure

Add cross-referenced actions requiring that:

- Protected- and conserved-area management effectiveness assessments include pesticide and fertiliser policies, HHP status, use records, sensitive-area buffers and non-target monitoring.
- Strategic water source areas, wetlands, riparian corridors, estuaries and freshwater priority areas are designated priority pesticide- and nutrient-monitoring zones.
- Environmental authorisations and management plans for agriculture, forestry, urban development and infrastructure in biodiversity priority areas include enforceable pesticide and nutrient conditions.
- Restoration finance, biodiversity credits, water tariffs and other nature-finance mechanisms do not fund HHP use and disclose all chemical inputs and ecological safeguards.

6. Required policy integration

The NBSAP should operate as the coordinating instrument that connects existing mandates rather than leaving pesticide and nutrient harms in separate institutional silos. The minimum integration required is set out below.

System	Primary instruments	Required NBSAP connection
Pesticide and fertiliser regulation	Act 36 of 1947; Pesticide Policy Framework and HHP policy process; One Health coordination	Use biodiversity and water evidence in registration, re-evaluation, withdrawal and substitution; publish an HHP schedule, registry and use data; accelerate biological alternatives.
Biodiversity and environmental governance	NEMA; NEMBA; White Paper on Conservation and Sustainable Use; National Biodiversity Framework	Apply precaution, prevention, polluter-pays, cooperative governance and public participation; make Target 7 outcomes enforceable through the NBF and implementation plans.
Alien species and restoration	NISSAP 2026-2036; Alien and Invasive Species Regulations; Working for Water; protected-area plans	Adopt non-chemical-first minimum standards, accelerated HHP exclusion, use reporting, sensitive-site buffers and ecological-recovery indicators.

System	Primary instruments	Required NBSAP connection
Water quality	National Water Act; NWRS-3; Integrated Water Quality Management Policy; National Eutrophication Management Strategy; WRC-DWS National Water Programme on Contaminants of Emerging Concern	Join pesticide and nutrient source-control to catchment plans, strategic water source areas, groundwater, event-based monitoring and ecological thresholds, CEC source control, drinking-water-source protection, and effects-based monitoring.
Wastewater and coastal discharges	National Water Act and applicable discharge authorisations; Water Services Act; Green Drop; NEMICMA coastal waters discharge permits; coastal-effluent guidance; municipal trade-effluent by-laws	Establish joint DWS–DFFE source-to-sea governance; establish dynamic, risk-based priority and watch lists and permit conditions; apply pollution prevention, source control, industrial pre-treatment, pharmaceutical take-back and risk-based treatment upgrades; targeted chemical and effects-based monitoring; monitor receiving ecosystems and whole-effluent toxicity; publish facility-level results and enforcement action.
Agriculture and land	CARA; Preservation and Development of Agricultural Land Act; provincial agricultural sector plans	Define sustainable agriculture through pesticide-risk and nutrient-loss outcomes; support agroecology, IPM, soil health and biological inputs while protecting food security and livelihoods.
Municipal and urban management	Municipal Systems Act; IDPs; biodiversity by-laws; stormwater, parks and procurement policies	Require pesticide-reduction plans, public notification and disclosure, HHP-free public spaces, manual and ecological management budgets and sensitive-area protections.
Forestry	National Forests Act; plantation management and certification systems; biodiversity plans	Track pesticide risk and HHP use independently of certification; require transition plans, riparian buffers, biological alternatives and transparent reporting.
Road and rail corridor management	SANRAL and National Roads Act 7 of 1998; SANRAL routine-road-maintenance specifications and EMPs; White Paper on National Rail Policy; Legal Succession to the South African Transport Services Act 9 of 1989; Railway Safety Act 30 of 2024; applicable TRIM, Transnet and PRASA vegetation-control specifications; NEMBA and Alien and Invasive Species Regulations; National Water Act; Act 36 and SANS 10206	Apply integrated, non-chemical-first vegetation management; distinguish safety-critical and ecological zones; phase out HHPs; protect sensitive biomes, water and communities; bind concessionaires and contractors; publish pesticide-use and incident data; and measure ecological condition rather than kilometres or hectares chemically treated.
Wildlife poisoning and human-wildlife conflict	National Vulture Multi-Species Biodiversity Management Plan; National Wildlife Poisoning Prevention Implementation Plan; NEMBA, TOPS and provincial conservation legislation; Act 36	Connect poisoning incidents and tissue-residue evidence to pesticide registration and use review, product-access controls, coordinated forensic response and enforcement; support non-lethal livestock-predator conflict measures; and include secondary rodenticide exposure in national pesticide-risk reporting.

7. Definitions and safeguards

7.1 Highly hazardous pesticides

The NBSAP should adopt the FAO/WHO concept of HHPs and require a transparent national list based on internationally recognised acute and chronic human-health and environmental hazard criteria, relevant international conventions and severe incident evidence. Environmental criteria must not be subordinated to proof of widespread poisoning or ecological collapse.

7.2 Safer alternatives and substitution

'Safer alternative' should mean an alternative shown, across its life cycle and full formulation, to present materially lower risks to human, animal, plant and ecosystem health and to be compatible with biodiversity recovery. Substitution decisions should compare non-chemical and biological options before chemical alternatives and should consider cumulative, mixture, resistance, drift, runoff, persistence and breakdown-product risks.

7.3 Integrated pest management

IPM must not be used as a label for routine pesticide programmes. It should require pest monitoring, prevention, thresholds, habitat and cultural management, natural enemy conservation, mechanical and biological controls, documented alternatives assessment and chemical use only as a last resort. HHP use is not compatible with an HHP phase-out pathway merely because a programme is described as IPM.

7.4 Nutrient loss

The nutrient target should measure nitrogen and phosphorus lost to the environment, not only fertiliser sales or application. Indicators should combine farm and catchment nutrient balances with nitrate and phosphate in ground and surface water, reactive nitrogen losses, nitrogen deposition, coastal eutrophication, soil health and nutrient-use efficiency.

7.5 Essential-use exemptions

Any exemption from an HHP restriction should be publicly notified; limited to a specific active ingredient, use, site and period; supported by evidence that no feasible safer option exists; accompanied by a funded substitution plan; independently reviewed; and automatically expire unless renewed through the same transparent process. Commercial inconvenience or cost alone should not justify an indefinite derogation.

7.6 Contaminants of emerging concern and other priority toxic pollutants

Contaminants of emerging concern are substances, particles or biological determinants - natural or synthetic, newly developed or long present - that are not routinely monitored or adequately regulated and whose occurrence, fate, mixture effects or human-health and ecological impacts require priority assessment. They include personal-care products, pharmaceuticals, their metabolites and transformation products; endocrine-disrupting chemicals; PFAS; antibiotics, resistant organisms and antimicrobial-resistance genes; micro- and nanoplastics; pesticides; industrial chemicals; and other trace organic and inorganic pollutants. National priority and watch lists must remain dynamic and risk-based.

Endocrine-disrupting chemicals are defined by their capacity to interfere with endocrine systems and may occur across several of these categories. HHPs and toxic metals are established priority pollutants and retain their dedicated risk-reduction, control and phase-out obligations; including them within integrated water monitoring must not dilute those obligations.

8. Governance, implementation and participation

- Establish a Target 7 interdepartmental implementation mechanism co-led by DFFE and the Department of Agriculture, with DWS, Health, Employment and Labour, Department of Transport, SANRAL, provincial road authorities, Transnet Rail Infrastructure Manager, PRASA, COGTA/SALGA, SANBI, SANParks, provincial authorities and the relevant research bodies.
- Include independent civil society, conservation NGOs, farm-worker and affected-community representatives, municipal workers and practitioners, CEC, environmental-chemistry and

ecotoxicology expertise, agroecology and biological-control expertise, and public-interest scientists. Conflicts of interest must be declared and managed.

- Affected wildlife cannot report exposure or participate in regulatory processes, therefore the implementation mechanism must include independent wildlife-ecotoxicology and species-conservation expertise, including organisations holding poisoning, rehabilitation, tissue-residue and mortality data on raptors, owls, scavengers and mammalian carnivores. This expertise must participate in indicator design, annual Target 7 review, referrals of wildlife evidence into Act 36 registration and re-evaluation decisions, and the development of public-sector procurement and pest-management standards.
- Publish an annual Target 7 dashboard covering pesticide risk, HHP withdrawals and exemptions, environmental concentrations, nutrient loss, transition finance, alternative uptake, incidents, enforcement and biodiversity outcomes.
- Use a regionally differentiated approach that identifies chemical and nutrient pressure hotspots in fragile and highly endemic systems, including the Cape Floristic Region, Succulent Karoo, Grassland Biome, Indian Ocean Coastal Belt, strategic water source areas, wetlands and estuaries.
- Provide a national complaint and incident-reporting pathway linked to environmental, water, labour and health authorities, with feedback to affected communities and anonymised public reporting.
- Review progress annually and formally in 2028 and 2030, with corrective measures where the trajectory is insufficient to meet the 2030 targets.
- Add “DFFE Oceans and Coasts, SAHPRA, municipal wastewater practitioners and water-pollution scientists” to the implementation mechanism.
- Expand the annual Target 7 dashboard to cover wastewater-system performance, spills and bypasses, Green Drop results, discharge-authorisation compliance, CEC and other priority-toxic-pollutant source control and monitoring, concentrations and loads, effects-based toxicity, treatment performance, corrective actions and enforcement.
- Add a formal 2028 review of Green Drop criteria, discharge authorisations, coastal permit conditions, municipal by-laws and receiving-water objectives.

9. Conclusion

South Africa cannot conserve biodiversity in protected islands while hazardous pesticides, excess nutrients, untreated or inadequately treated wastewater and persistent synthetic pollutants continue to move through farms, forests, cities, catchments and conservation programmes into rivers, wetlands, groundwater, estuaries and coastal waters. The NBSAP is the rare national instrument capable of connecting these pressures into one coherent prevention, transition and accountability agenda.

The draft has made an important start by recognising pollution, pesticides and the need for improved water-quality governance. It will only fulfil that promise if it restores the quantitative 2030 commitments; adopts a transparent HHP phase-out pathway; governs fertiliser losses; reaches every pesticide-use sector; prevents chemical harm in alien-species control and restoration work; and makes progress visible through public data and biodiversity outcomes.

It must also close the gaps in wastewater and coastal-discharge governance. This requires expressly assigning DWS responsibility alongside DFFE Oceans and Coasts, COGTA/SALGA and municipalities; strengthening Green Drop criteria and linked discharge authorisations; establishing enforceable effluent, receiving-water and ecological requirements; and introducing risk-based source control, targeted and effects-based monitoring and public reporting for contaminants of emerging

concern and other priority toxic pollutants, including endocrine-disrupting chemicals, pharmaceuticals and transformation products, PFAS, micro- and nanoplastics and antimicrobial-resistance determinants, resistant organisms and antimicrobial-resistance genes.

UnPoison requests incorporation of these amendments before the NBSAP is finalised and submitted to Cabinet and the Convention on Biological Diversity.



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Supporting Organisations



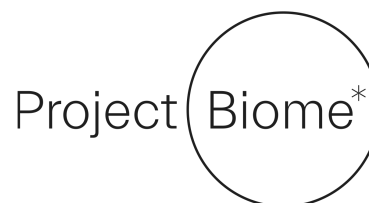
<https://biowatch.org.za/>



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